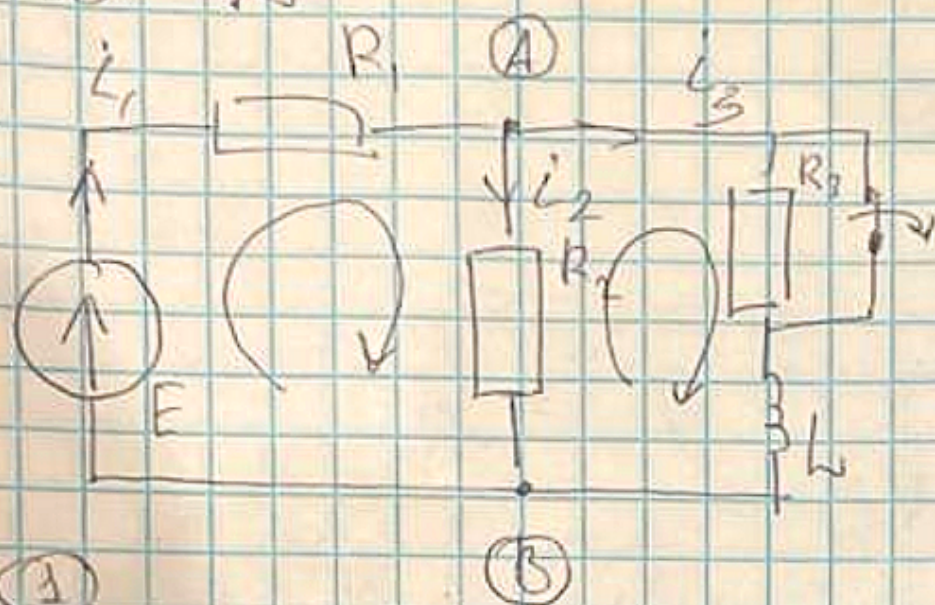


Компьютерная работа №2

В-18



①

③

$$\begin{cases} i_1 - i_2 - i_3 = 0 \\ I_1 R_1 + I_2 R_2 = E \\ I_3 R_3 - I_2 R_2 + U_L = 0 \end{cases}$$

$$U_L = L \frac{dI_3}{dt}$$

$$\begin{bmatrix} 1 & -1 & -1 \\ R_1 & R_2 & 0 \\ 0 & -R_2 & R_3 + sL \end{bmatrix} = 0$$

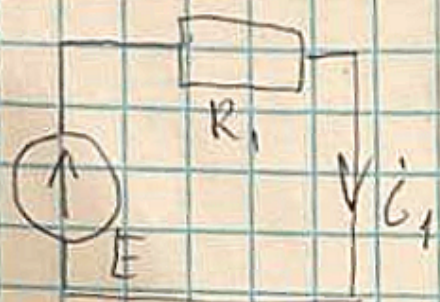
$$\begin{bmatrix} 1 & -1 & -1 \\ 100 & 100 & 0 \\ 0 & -100 & 100 + 0.5s \end{bmatrix} = 0$$

$$10^4 + 50L + 10^4 + 10^4 + 50L = 0$$

$$100L = -30000$$

$$L = -300$$

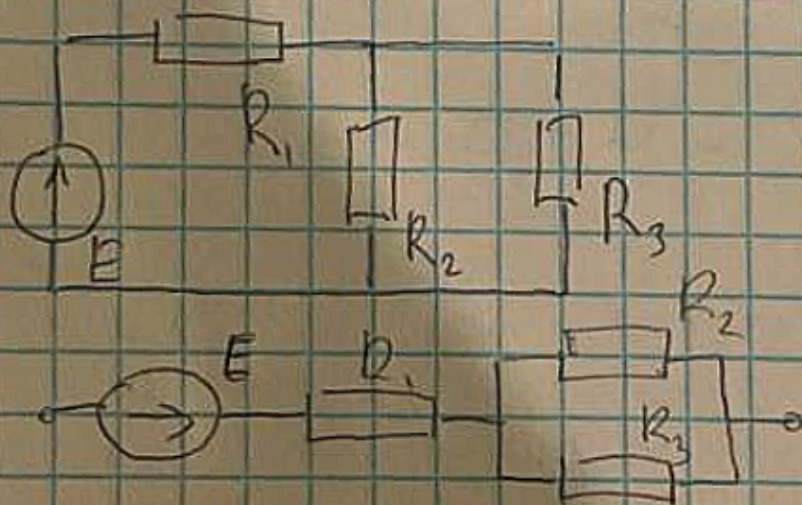
② $t < 0$



$$I_L = \frac{E}{R_1}$$

$$I_L = 0,3(A)$$

③ $t \rightarrow \infty$



$$R_{\text{eq}} = R_1 + \frac{R_2 \cdot R_3}{R_2 + R_3} = 150(\Omega)$$

$$I_{\text{eq}} = \frac{E}{R_{\text{eq}}} = 0,2(A)$$

$$I_{30} = \frac{I_{\text{eq}}}{2} = 0,1(A)$$

$$I_3(t) = 0,1 + (0,3 - 0,1)e^{-300t}$$

